Ultramicroscopy 177 (2017) 78–83

Contents lists available at ScienceDirect

Ultramicroscopy

journal homepage: www.elsevier.com/locate/ultramic

Streamlined approach to mapping the magnetic induction of skyrmionic materials

Jordan J. Chess[a,*], Sergio A. Montoya[b,c], Tyler R. Harvey[a], Colin Ophus[d], Simon Couture[b,c], Vitaliy Lomakin[b,c], Eric E. Fullerton[b,c], Benjamin J. McMorran[a]

[a] Department of Physics, University of Oregon, Eugene, OR 97403, USA
[b] Center for Memory and Recording Research, University of California, San Diego, CA 92093, USA
[c] Department of Electrical and Computer Engineering, University of California, San Diego, La Jolla, CA 92093, USA
[d] National Center for Electron Microscopy, Molecular Foundry, Lawrence Berkeley National Laboratory, 1 Cyclotron Road, Berkeley, CA 94720, USA

ARTICLE INFO

Article history:
Received 19 October 2016
Revised 13 February 2017
Accepted 22 February 2017
Available online 28 February 2017

ABSTRACT

Recently, Lorentz transmission electron microscopy (LTEM) has helped researchers advance the emerging field of magnetic skyrmions. These magnetic quasi-particles, composed of topologically non-trivial magnetization textures, have a large potential for application as information carriers in low-power memory and logic devices. LTEM is one of a very few techniques for direct, real-space imaging of magnetic features at the nanoscale. For Fresnel-contrast LTEM, the transport of intensity equation (TIE) is the tool of choice for quantitative reconstruction of the local magnetic induction through the sample thickness. Typically, this analysis requires collection of at least three images. Here, we show that for uniform, thin, magnetic films, which includes many skyrmionic samples, the magnetic induction can be quantitatively determined from a single defocused image using a simplified TIE approach.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Magnetic skyrmions are particle-like spin textures that possess nontrivial topology [1]. The stability of skyrmions and the low current density necessary to move them [2] has inspired many suggested applications that employ skyrmions as bits in both memory and logic devices which are predicted to be highly energy-efficient [3–8]. These magnetic quasi-particles were initially identified only at low temperatures in non-centrosymmetric crystals including MnSi [9,10], FeCoSi [11] and FeGe [12], but recent observations have shown that skyrmions can be stabilized in a more diverse class of materials including those with perpendicular magnetic anisotropy (PMA) [8,13–17]. This larger swath of materials suggests the need for more rapid characterization techniques to both facilitate the efficient search for materials suitable for applications in skyrmionic devices and explore the basic physics of these magnetic textures.

Lorentz transmission electron microscopy (LTEM) is one of a very few techniques that provides direct real space images of magnetic features at the nanoscale. Recent improvements in aberration correction and instrument stability have led to a new resolution benchmark of 1 nm for scanning LTEM [18]. Additionally, new tomographic reconstruction algorithms have led to the demonstration of 3D vector field electron tomography by Phatak et al. [19]

Most of the LTEM studies of skyrmion materials have employed analysis based on the transport of intensity equation (TIE), [20,21] an equation that relates the z-derivative of the image intensity to the phase shift of an electron. This approach yields quantitative maps of the local in-plane magnetic induction integrated through the sample thickness, but requires multiple images (under-, in-, and over-focused) be taken at a specific point of interest in the sample [21]. In a post-processing step these images are first aligned and then used to approximate the z-derivative of the image intensity. In order to maximize the final field of view, the microscopist must carefully align the microscope to minimize image movement between images recorded at different focus values. These alignments can be sensitive to changes in other experimental parameters including magnetic field applied to the sample. This physical alignment, coupled with the need to properly align images, which can be difficult to automate [22], increases the total time needed to extract useful information from a magnetic sample. This often makes certain experiments prohibitively time-consuming, such as determining the in-plane magnetic induction during an *in-situ* applied field sweep (although this type of study does exists in the LTEM literature [23,24]). One approach, differential phase imaging [25], developed by Pollard et al., maps the

* Corresponding author.
E-mail address: jchess@uoregon.edu (J.J. Chess).

http://dx.doi.org/10.1016/j.ultramic.2017.02.004
0304-3991/© 2017 Elsevier B.V. All rights reserved.

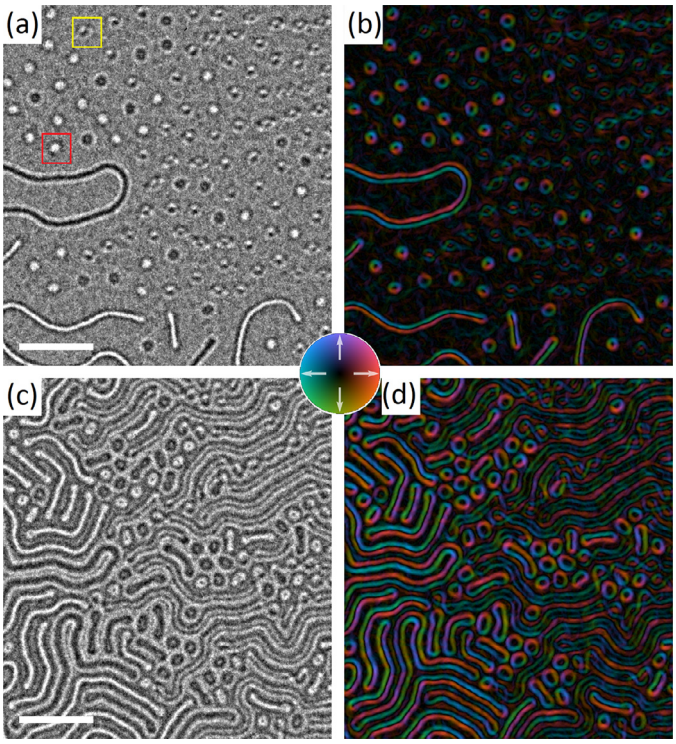

Fig. 1. (a) and (c) Selected under-focused Lorentz TEM images from a field sweep performed on a Fe/Gd multilayered thin film with (a, b) 180 mT and (c, d) 70 mT field applied perpendicular to the film. Scale bar is 1 μm. The red and yellow square in (a) highlight a skyrmion and topologically trivial magnetic bubble respectively. (b), (d), The magnetic induction calculated using our single image analysis on the image to the left (hue and saturation of color indicate the direction and magnitude of the magnetic induction). See supplemental material video for full field sweep. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

change in magnetic state during a dynamic measurement. This approach can be applied to non-uniform films. Alternatively, one can forego mapping the magnetic induction, and instead answer questions that depend only on the location of domain walls, which can in general be accomplished with a single defocused image. This method has been used to determine the non-adiabatic spin torque parameter [26], image domain wall nucleation [27], and record skyrmion motion [28]. Additionally, Phatak et al., showed that both the polarity and chirality of a vortex magnetization pattern of a magnetic disk can be determined from a single Fresnel contrast image of a tilted sample [29].

Similar to the work by Eastwood et al. [30] or Koch [31], in which they present iterative algorithms for single image phase reconstruction, here we show that one defocused image is sufficient to determine the magnetic portion of the electron phase shift of a uniform film using a simplified TIE approach. This allows one to map the magnetic induction without the trade-off of a slower, more involved focal series experiment, making it ideal for *in-situ* experiments on suitable samples. Fig. 1 shows an application of the single image TIE approach we are discussing here, applied to an [Fe0.36nm/Gd0.4nm] × 80 multilayered film [16,32], under quasi-dynamic conditions. The data was taken as an applied perpendicular magnetic field was swept from a field strong enough to saturate the sample to a slightly negative applied field. The data shows dipole skyrmions (black/white circles (red square in Fig. 1 (a))), disordered stripe domains, and bubbles with zero topological charge (elliptically shaped (yellow square in Fig. 1 (a))) nucleating as the field strength is reduced. These features then evolve during the field sweep into a mixture of skyrmions and labyrinth domains. The top two images (a,b) are the under-focus LTEM image and reconstructed magnetic induction with $\Delta f = -300$ μm, and applied field $H_z = 180$ mT, while (c) and (d) are the under-focus and magnetic induction at $H_z = 70$ mT.

Additional algorithms for single-image phase retrieval or exit-wave reconstruction exist but require specific sample geometries such as an isolated object [33], or specific illumination conditions and a diffraction image [34,35] which make them not suitable for this type of sample or difficult to implement in a TEM. It is worth emphasizing that the general paradigm for these single-image phase retrieval algorithms is to use *a priori* knowledge to simplify the analysis, which in practice usually means restricting oneself to a subset of samples. In this case we are choosing to restrict our analysis to uniform, thin, magnetic films, which can be treated as pure magnetic phase objects. Utilizing our new approach, the full in-plane magnetic induction can be determined for each image in a quasi-dynamic measurement with no extra experimental requirements and fewer post-processing steps. This fuller understanding is often required to interpret the LTEM images of the complex magnetization textures present in skyrmionic materials.

2. Theory

The phase imparted on an electron plane wave traveling along the z-axis after transmission through a sample with electric potential V and vector potential $\mathbf{A}$ is given by the Aharonov–Bohm phase shift [36]:

$$\begin{aligned}\phi(\mathbf{r}_\perp) &= C_E \int_L V(\mathbf{r}_\perp, z)\mathrm{d}z - \frac{\pi}{\Phi_0}\int_L \mathbf{A}(\mathbf{r}_\perp, z)\cdot \mathrm{d}\mathbf{r} \\ &\equiv \phi_e + \phi_m,\end{aligned} \tag{1}$$

where L is a path parallel to the propagation direction of the electron beam, $\mathbf{r}_\perp$ is the location in the sample plane, C_E is the interaction constant [21], and Φ_0 is the magnetic flux quantum ($\frac{h}{2e}$). If we assume a homogeneous foil of uniform thickness d and constant mean inner potential (V_0) the electrostatic term can be easily evaluated and yields,

$$\phi_e = C_E V_0 d,$$

Additionally, the effects of inelastic scattering and high angle scattering of electrons out of the optical system can be described by an exponential drop in the initial amplitude of the electron wave function. Thus, assuming parallel illumination, the complex amplitude exiting the foil is,

$$\psi_0(\mathbf{r}_\perp) = Ae^{-\alpha d}e^{iC_E V_0 d}e^{i\phi_m(\mathbf{r}_\perp)}. \tag{2}$$

The intensity of the wave at the image plane using the microscope transfer function ($\mathcal{T}(\mathbf{q}_\perp)$) is then given by,

$$I(\mathbf{r}_\perp, \Delta f) = |\mathscr{F}^{-1}\{\mathscr{F}[\psi_0(\mathbf{r}_\perp)]\mathcal{T}(\mathbf{q}_\perp)\}|^2 \tag{3}$$

where $\mathbf{q}_\perp$ are the in-plane spatial frequencies. A relevant transfer function that models the effects of spherical aberration (C_s) and a damping envelope ($E_s(q_\perp)$) due to a spread in illumination angles caused by lens instabilities is:

$$\mathcal{T}(\mathbf{q}_\perp) = a(|\mathbf{q}_\perp|)e^{-i\chi(q_\perp)}e^{-E_s(q_\perp)}, \tag{4}$$

where $a(q_\perp)$ is an aperture function, the phase transfer function $\chi(q_\perp)$ is described by,

$$\chi(q_\perp) = \pi\lambda\Delta f q_\perp^2 + \frac{1}{2}\pi C_s\lambda^3 q_\perp^4, \tag{5}$$

and $E_s(q_\perp)$ given in terms of the divergence angle Θ_c is [37],

$$E_s(q_\perp) = \left(\frac{\pi\Theta_c}{\lambda}\right)^2\left(C_s\lambda^3 q_\perp^3 + \Delta f\lambda q_\perp\right)^2. \tag{6}$$

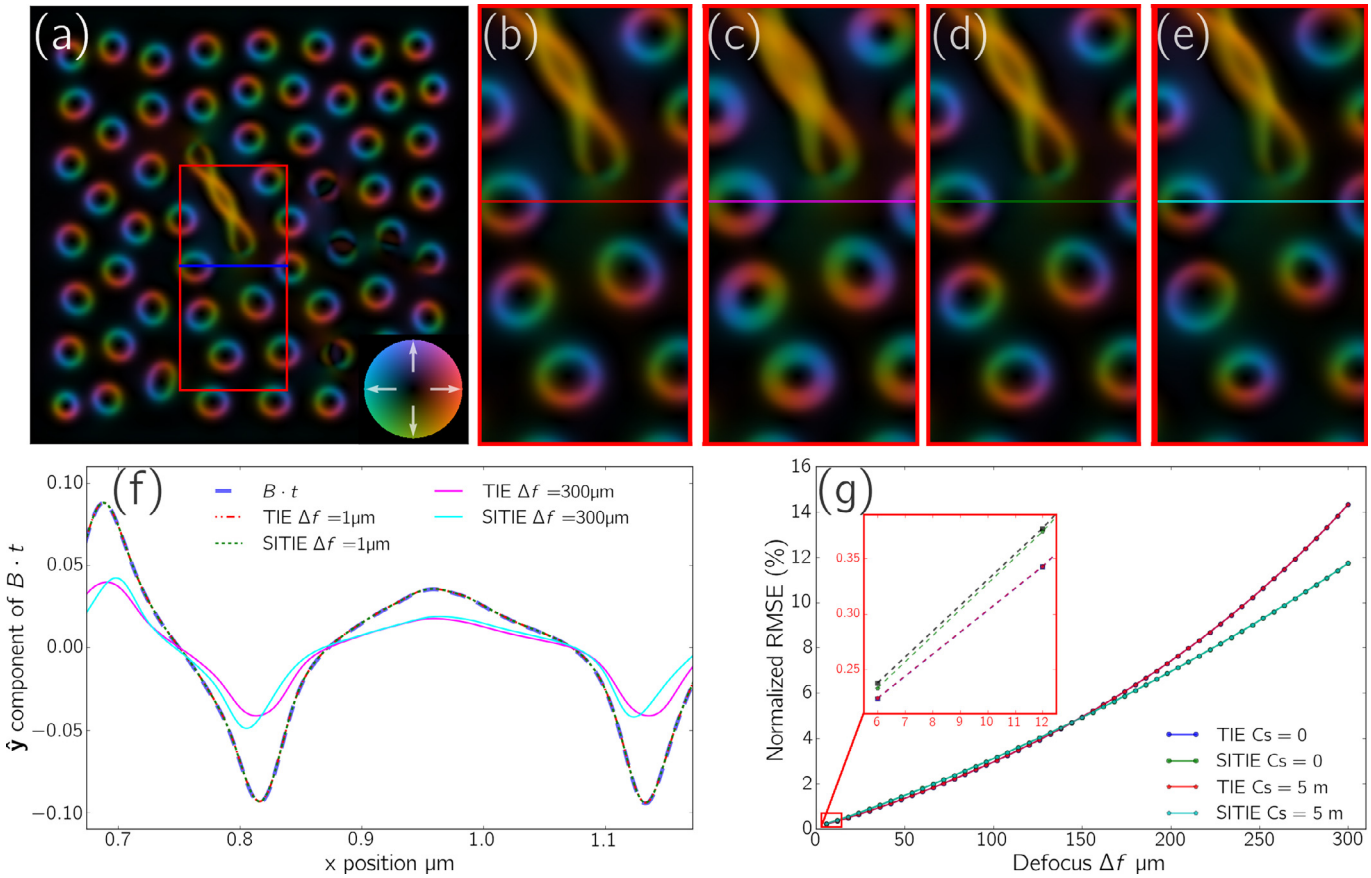

Fig. 2. Simulations (a) Plot of in-plane components of the local magnetic induction calculated from output of a micromagnetic simulation (color indicates direction and magnitude of the field). (b–e) Local magnetic induction of the region shown in the red box in (a), calculated using conventional TIE with $\Delta f = 1$ μm and $\Delta f = 300$ μm for (b) and (c) respectively. The red box has a 0.5 μm width. SITIE-calculated magnetic induction for (d) $\Delta f = 1$ μm and (e) $\Delta f = 300$ μm. Notice that only slight distortion errors are present in the $\Delta f = 300$ μm cases, caused by using a focus outside of the validity of the paraxial approximation. (f) y-component of the magnetic induction along the colored lines in images (a–e), note the nearly perfect agreement between the reference, TIE, and SITIE for the $\Delta f = 1$ μm. (g) Plot of the total normalized root mean square error in the determination of $B \cdot t$ calculated using Eq. (12) as function of defocus for TIE and SITIE showing there is no practical difference between the methods for moderate defocus. For these simulations Θ_c was set at 5×10^{-5} rad. The inset shows that for any Δf larger than 10 μm the effects of including a non-zero C_s are truly negligible. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Above, λ is the relativistic electron wavelength, Δf is the distance from the in-focus plane, and we have used $q_\perp \equiv |\mathbf{q}_\perp|$ for notational convenience. Before continuing we stop to note that conventional TIE analysis presumes that all terms higher in order than $q_\perp^2$ in Eqs. (5) and (6) are negligible. This is generally a reasonable assumption because of the large defocus values used in LTEM, for example see Fig. 2. As will be discussed, our method neglects one additional term (the last term in Eq. (6)). For completeness and accuracy the full transfer function (Eq. (4)) was used in all image simulations.

Taylor expanding the transfer function for small $q_\perp$, the "paraxial approximation", we arrive at an approximate form of Eq. (3) [38],

$$I(\mathbf{r}_\perp, \Delta f) \approx I_0 - \frac{\lambda \Delta f}{2\pi} \nabla_\perp \cdot (I_0 \nabla_\perp \phi_m) + \frac{(\pi \Theta_c \Delta f)^2}{2} [\sqrt{I_0} \nabla_\perp^2 \sqrt{I_0} - I_0 (\nabla_\perp \phi_m)^2], \quad (7)$$

where $I_0 = |\psi_0(\mathbf{r}_\perp)|^2$. Examining Eq. (2), we see that if we are analyzing homogeneous thin-film specimens with a uniform thickness, which includes many materials, then I_0 becomes a constant, as shown, for example, in Fig. 3b. We emphasize that this technique can only be applied to regions that have no amplitude or diffraction contrast in the in-focus image. The amorphous material shown in this paper meets this criteria, as do many magnetic thin film materials. Eq. (7) simplifies to,

$$I(\mathbf{r}_\perp, \Delta f) \approx I_0 \left(1 - \frac{\lambda \Delta f}{2\pi} \nabla_\perp^2 \phi_m - \frac{(\pi \Theta_c \Delta f)^2}{2} (\nabla_\perp \phi_m)^2 \right). \quad (8)$$

As show by De Graef et al. [38] the transport of intensity equation can be obtained from (8) by simply subtracting the value at $I(\mathbf{r}_\perp, \pm\Delta f)$ yielding,

$$\nabla_\perp^2 \phi_m = -\frac{2\pi}{I_0 \lambda} \frac{I(\mathbf{r}_\perp, \Delta f) - I(\mathbf{r}_\perp, -\Delta f)}{2\Delta f} \approx -\frac{2\pi}{I_0 \lambda} \frac{\partial I}{\partial z}. \quad (9)$$

In this way, the Laplacian of the phase can be derived from two different images of the specimen recorded under different focal conditions. Eq. (9) is the standard equation used in analyzing LTEM data. Note that a crucial step in standard use of TIE analysis is the calculation of the difference between two images (Eq. (9) RHS). Thus, the reconstructed magnetic phase is subject to errors introduced when acquiring images under different conditions including: drift, rotations, and changes in magnification.

Here we suggest that due to the high spatial coherence of modern field emission microscopes (Θ_c in the range (0.01–0.08) mrad [21,31,37]), the effects of the last term in Eq. (8) are negligible at the medium resolutions used for LTEM. We present Fig. 3 as evidence of the above statement. Additionally, the validity of this approximation and its implications are discussed in detail in the supplementary material as well as the effects of noise.

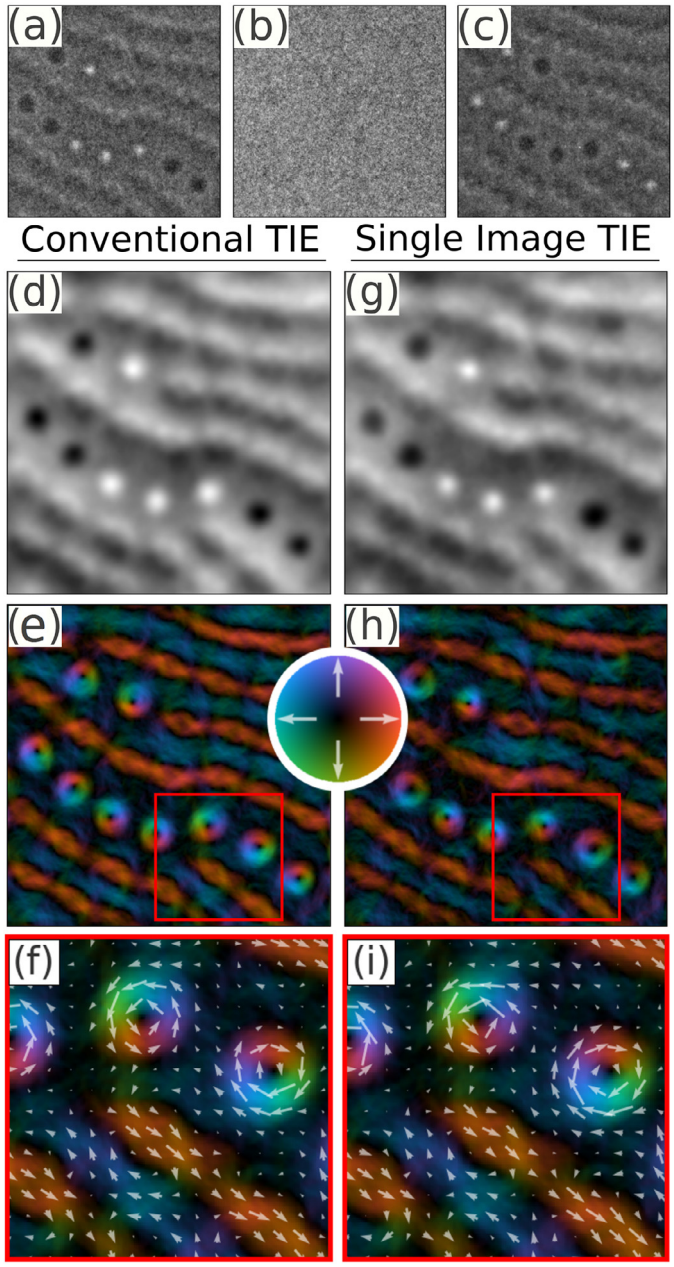

Fig. 3. (a–h) Experimental Lorentz TEM analysis of a Fe/Gd multilayered thin film over the same 1.5 µm field of view. (a) Under-focused, (b) in-focus, and (c) over-focused images showing Fresnel-contrast ((a,c) recorded at $\Delta f = \pm$ 300 µm). (d) Phase calculated using the standard TIE applied to image (a–c). (g) Phase calculated using only image (a). (e), (h), The magnetic induction calculated from phase above. (f), (i) Enlarged area from boxed region in (e), (h) with magnetization represented both by color and vector arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This simplification results in a Single Image Transport of Intensity Equation (SITIE),

$$\nabla_{\perp}^{2}\phi_m \approx \frac{2\pi}{\lambda\Delta f}\left(1 - \frac{I(\mathbf{r}_{\perp}, \Delta f)}{I_0}\right). \tag{10}$$

One then needs to determine a value for I_0, which is given by the average of the out-of-focus image due to the preservation of electron flux [31]. Alternatively, one can arrive at Eq. (10) by simply starting with the TIE equation, assuming the object in question is a pure phase object, and using $I(\mathbf{r}_{\perp}, \Delta f) - I_0$ as a $\mathcal{O}(\Delta f)$ approximation to $\partial I/\partial z$. Here we chose to present a more involved derivation to make clear what was being neglected when making this first-order approximation of the derivative, when compared to the more traditional $\mathcal{O}(\Delta f^2)$ approach used in Eq. (9).

Multiple techniques have been developed to solve the standard TIE equation including a Fourier-based approach [39], a multigrid algorithm [40], a symmetrized version of the Fourier method [41], and finite element method [42], all of which can also be applied to the SITIE to determine the phase of the exit wave. From this phase, the local magnetic induction can easily be determined using the relation,

$$\nabla_{\perp}\phi_m(\mathbf{r}_{\perp}) = -\frac{e}{\hbar}[\mathbf{B}(\mathbf{r}_{\perp}) \times \hat{\mathbf{e}}_{\mathbf{z}}]d, \tag{11}$$

where $\hat{\mathbf{e}}_{\mathbf{z}}$ is a vector parallel to the beam propagation direction.

3. Methods

3.1. Micromagnetic simulations

To validate SITIE and quantify the errors associated with this method, we simulated through-focal series images of exactly known, simulated magnetization textures. These micromagnetic textures were obtained from Landau–Lifshitz–Gilbert simulations calculated using the FastMag solver [43]. The micromagnetic simulation is for a 2 µm × 2 µm × 80 nm ferromagnetic film, using experimentally measured values for the saturation magnetization (M_s = 0.4 A/µm), anisotropy constant (K = 4 × 10^4J/m^3), Gilbert damping ($\alpha = 0.05$), and exchange stiffness (A_{ex} = 5 × 10^5 J/m). An applied perpendicular magnetic field of H_z = 0.2 T was used, and the system is allowed to relax to an equilibrium state in 10 ns. These parameters result in the in-plane magnetic induction pattern shown in Fig. 2a.

3.2. Lorentz Image Simulations

Fresnel-contrast LTEM images were simulated using the Mansuripur algorithm: the magnetic phase shift imparted on the electron wave by the results of the micromagnetic simulation was calculated, and then Eq. (4) was used to propagate the wave to a given defocus plane [44]. The electrostatic phase shift was neglected in the simulations, in line with the theory above, as it only contributes an overall constant phase and doesn't contribute to the image intensity. Prior to applying the Mansuripur algorithm, the output of the FastMag simulations were expanded from 200 × 200 arrays to 2048 × 2048, and then padded with zeros to a total array size of 4096 × 4096 to mitigate the introduction of any artifacts from the Fourier-based approach used in both the Mansuripur algorithm and transfer function formalism. The parameters used for image simulations were: accelerating voltage 300 kV, defocus values Δf = 1 µm–300 µm, and spherical aberration C_s = 0–5 m. These values more than cover the range encountered in both standard and aberration-corrected microscopes during an LTEM experiment. The normalized root mean square error is used as an metric to compare the reconstructed phase to the known phase, calculated as,

$$\mathbf{NRMSE} = \frac{\sum_{i=0}^{i=1}\sqrt{\frac{\sum_{m,n}\left((\tilde{\mathbf{B}}t)_{n,m} - (\mathbf{B}t)_{n,m}\right)^2}{nm}}}{(\mathbf{B}t)_{\max} - (\mathbf{B}t)_{\min}} \times 100\% \tag{12}$$

where $\tilde{\mathbf{B}}$ is the TIE/SITIE reconstructed local magnetic induction, $\mathbf{B}$ is the known magnetic induction, t the sample thickness, (m, n) the array indices, and $(i = 0, 1)$ the components of the vector.

4. Evaluation of SITIE

4.1. Numerical evaluation

To quantitatively analyze the validity of SITIE compared to TIE, we numerically simulated Fresnel-contrast images from simulated domain structures obtained from the micromagnetic simulation. This allows us to compare the two phase retrieval methods in the absence of noise or any misalignments in images that could cause errors in standard TIE analysis. Additionally, it gives us a known reference to quantify results that is not present when analyzing experimental data. The Fourier transform-based method of solving the transport of intensity equation was utilized to reconstruct the phase of both the experimental and simulated data [21,45]. A comparison of the two methods applied to experimental data is left to the next Section 4.2.

Fig. 2(b–e), show the calculated local magnetic induction from both TIE (b,c) and SITIE (d,e) each under two different focal conditions; the first (b,d) from a small defocus (1 µm) and the second from a large defocus value (c,e) (300 µm). Notice the close agreement between the reference and both TIE and SITIE for small defocus (Fig. 2a,b,d), which have a normalized root mean square error (NRMSE) of 0.169% and 0.170% respectively. Interestingly, for the large defocus (300 µm) examples (Fig. 2e,f) the error associated with TIE (14.3%) is larger than that for SITIE (11.7%). These results can be understood analytically from the right hand side of Eqs. (9) and (10). They are the central and forward difference approximations for the z-derivative of the image intensity, and have associated errors of order $\mathcal{O}(\Delta f^2)$ and $\mathcal{O}(\Delta f)$ respectively. This quadratic versus linear error is evident in Fig. 2g. Also, evident in Fig. 2g is the well-known fact that for all but the smallest defocus values used in LTEM, the effects of spherical aberration are negligible [21].

4.2. Experimental evaluation

To ensure the validity of SITIE on real data, we collected and analyzed through-focal series images of magnetic bubble domains in a thin film sample. The images were collected using an FEI Titan equipped with a Lorentz lens and integrated CEOS objective lens image aberration corrector. The standard objective lens was partially excited to apply a magnetic field perpendicular to the sample plane. The sample is nominally a [Fe 0.34nm/ Gd 0.4nm] $\times$ 80 multilayered film deposited by DC magnetron sputtering onto 50 nm Si_3N_4 membrane with 20 nm Ta seed and capping layers [16].

Prior to analysis all experimental images were filtered following the method suggested by Tasdizen et al. to remove low-frequency artifacts caused by slightly non-uniform illumination [46]. Fig. 3 shows the focal series (a–c) for $\Delta f = (-300,0,300$ µm). The left column shows (d) the phase reconstructed using conventional TIE analysis applied to (a–c), (e) the magnetic induction determined using the phase in (d) represented with color indicating the magnitude and direction of the magnetic induction and (f) a closer look at the region inside the red square in (e). The right column shows the phase (g) and magnetic induction (h,i) determined using only image (a). Included in the images are skyrmions, four of which have helicity $\gamma = \pi/2$ (white circles in phase images), and five with $\gamma = -\pi/2$, with γ defined to be consistent with equation B3 in [1]. The remaining features are stripe domains which are starting to break up into topologically trivial bubbles, and skyrmion bound pairs [16]. It is important to note that the slightly lower signal-to-noise present in Fig. 3g is not an inherent difference between SITIE and TIE, but instead a consequence of Fig. 3g having half the effective exposure time due to it being calculated from only one image. This difference could be easily accounted for by increasing the exposure time for images collected for SITIE, or by collecting multiple shorter-exposure images aligning and averaging them later. We emphasize here that aligning images collected at the same focus value can be accomplished using simple algorithms such as cross-correlation and is significantly easier than aligning images at different foci, because of the associated reversals in contrast, rotation, and distortions between images. Errors in image alignment caused by pixel shift, magnification changes, and rotations can cause significant errors in the reconstructed phase when performing TIE analysis. For a detailed discussion of this subject we refer the reader to chapter 5.3.2 of De Graef and Zhu [21]. SITIE is free of all these errors.

5. Conclusion

We have demonstrated, both numerically and experimentally, that a single Lorentz TEM image can appropriately be used to map the magnetic phase of uniform samples, specifically for thin films exhibiting skyrmionic phases. This simplified TIE approach gives roughly equivalent results to conventional TIE analysis. Using SITIE analysis on uniform samples simplifies both the computational load and data collection involved in characterizing topological magnetization textures. Furthermore, this simplification opens the door to exploring new phenomenon that was previously impractical with the traditional TIE analysis by removing the need to align and collect multiple images, and reducing errors caused by distortions in images. This simplified technique allows for phase reconstruction during quasi-dynamic measurements (e.g. field and/or temperature sweeps), and Simplifies the route to ultra-fast LTEM studies [47].

Acknowledgments

This work was partially supported by the U.S. Department of Energy, Office of Science, Basic Energy Sciences, under Award No. DE-SC0010466 (JJC, TRH, BJM). CO acknowledges that work at the Molecular Foundry was supported by the Office of Science, Office of Basic Energy Sciences, of the U.S. Department of Energy under Contract No. DE-AC02- 05CH11231. Work at UCSD including materials synthesis and characterization was supported by U.S. Department of Energy (DOE), Office of Basic Energy Sciences (Award No. DE-SC0003678). The authors wish to thank Josh Razink of the CAMCOR facility at University of Oregon. The CAMCOR High-Resolution and Nanofabrication Facility (TEM, FIB and SEM) are supported by grants from the W.M Keck Foundation, the M.J. Murdock Charitable Trust, ONAMI, the Air Force Research Laboratory (agreement number FA8650–05–1–5041), NSF (award numbers 0923577, 0421086) and the University of Oregon.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.ultramic.2017.02.004.

References

[1] N. Nagaosa, Y. Tokura, Topological properties and dynamics of magnetic skyrmions, Nat. Nanotechnol. 8 (12) (2013) 899–911, doi:10.1038/nnano.2013.243.
[2] G. Yu, P. Upadhyaya, X. Li, W. Li, S.K. Kim, Y. Fan, K.L. Wong, Y. Tserkovnyak, P.K. Amiri, K.L. Wang, Room-temperature creation and spin orbit torque manipulation of skyrmions in thin films with engineered asymmetry, Nano Lett. (2016), doi:10.1021/acs.nanolett.5b05257.
[3] R. Tomasello, E. Martinez, R. Zivieri, L. Torres, M. Carpentieri, G. Finocchio, A strategy for the design of skyrmion racetrack memories, Sci. Rep. 4 (2014) 00003, doi:10.1038/srep06784.
[4] X. Zhang, Y. Zhou, M. Ezawa, G.P. Zhao, W. Zhao, Magnetic skyrmion transistor: skyrmion motion in a voltage-gated nanotrack, Sci. Rep. 5 (2015) 11369, doi:10.1038/srep11369.

[5] X. Zhang, G.P. Zhao, H. Fangohr, J.P. Liu, W.X. Xia, J. Xia, F.J. Morvan, Skyrmion-skyrmion and skyrmion-edge repulsions in skyrmion-based racetrack memory, Sci. Rep. 5 (2015) 7643, doi:10.1038/srep07643.

[6] S. Krause, R. Wiesendanger, Spintronics: skyrmionics gets hot, Nat. Mater. 15 (5) (2016) 493–494, doi:10.1038/nmat4615.

[7] X. Zhang, M. Ezawa, Y. Zhou, Thermally stable magnetic skyrmions in multilayer synthetic antiferromagnetic racetracks, arXiv:1601.03893 [cond-mat] (2016). ArXiv: 1601.03893.

[8] S. Woo, K. Litzius, B. Krüger, M.-Y. Im, L. Caretta, K. Richter, M. Mann, A. Krone, R.M. Reeve, M. Weigand, P. Agrawal, I. Lemesh, M.-A. Mawass, P. Fischer, M. Kläui, G.S.D. Beach, Observation of room-temperature magnetic skyrmions and their current-driven dynamics in ultrathin metallic ferromagnets, Nat. Mater. advance online publication (2016), doi:10.1038/nmat4593.

[9] S. Mühlbauer, B. Binz, F. Jonietz, C. Pfleiderer, A. Rosch, A. Neubauer, R. Georgii, P. Böni, Skyrmion lattice in a chiral magnet, Science 323 (5916) (2009) 915–919, doi:10.1126/science.1166767.

[10] X.Z. Yu, Y. Onose, N. Kanazawa, J.H. Park, J.H. Han, Y. Matsui, N. Nagaosa, Y. Tokura, Real-space observation of a two-dimensional skyrmion crystal, Nature 465 (7300) (2010) 901–904, doi:10.1038/nature09124.

[11] W. Munzer, A. Neubauer, T. Adams, S. Muhlbauer, C. Franz, F. Jonietz, R. Georgii, P. Boni, B. Pedersen, M. Schmidt, A. Rosch, C. Pfleiderer, Skyrmion lattice in the doped semiconductor Fe1-x Cox Si, Phys. Rev. B 81 (4) (2010) 041203, doi:10.1103/PhysRevB.81.041203.

[12] X.Z. Yu, N. Kanazawa, Y. Onose, K. Kimoto, W.Z. Zhang, S. Ishiwata, Y. Matsui, Y. Tokura, Near room-temperature formation of a skyrmion crystal in thin-films of the helimagnet FeGe, Nat. Mater. 10 (2) (2011) 106–109, doi:10.1038/nmat2916.

[13] X. Yu, M. Mostovoy, Y. Tokunaga, W. Zhang, K. Kimoto, Y. Matsui, Y. Kaneko, N. Nagaosa, Y. Tokura, Magnetic stripes and skyrmions with helicity reversals, Proc. Nat. Acad. Sci. 109 (23) (2012) 8856–8860, doi:10.1073/pnas.1118496109.

[14] X.Z. Yu, Y. Tokunaga, Y. Kaneko, W.Z. Zhang, K. Kimoto, Y. Matsui, Y. Taguchi, Y. Tokura, Biskyrmion states and their current-driven motion in a layered manganite, Nat. Commun. 5 (2014) 3198, doi:10.1038/ncomms4198.

[15] D. Morikawa, X.Z. Yu, Y. Kaneko, Y. Tokunaga, T. Nagai, K. Kimoto, T. Arima, Y. Tokura, Lorentz transmission electron microscopy on nanometric magnetic bubbles and skyrmions in bilayered manganites La1.2sr1.8(Mn1yRuy)2o7 with controlled magnetic anisotropy, Appl. Phys. Lett. 107 (21) (2015) 212401, doi:10.1063/1.4936078.

[16] J.C.T. Lee, J.J. Chess, S.A. Montoya, X. Shi, N. Tamura, S.K. Mishra, P. Fischer, B.J. McMorran, S.K. Sinha, E.E. Fullerton, S.D. Kevan, S. Roy, Synthesizing skyrmion bound pairs in Fe-Gd thin films, Appl. Phys. Lett. 109 (2) (2016) 022402, doi:10.1063/1.4955462.

[17] W. Wang, Y. Zhang, G. Xu, L. Peng, B. Ding, Y. Wang, Z. Hou, X. Zhang, X. Li, E. Liu, S. Wang, J. Cai, F. Wang, J. Li, F. Hu, G. Wu, B. Shen, X.-X. Zhang, A centrosymmetric hexagonal magnet with superstable biskyrmion magnetic nanodomains in a wide temperature range of 100-340 K, Adv. Mater. (2016). n/a–n/a. doi: 10.1002/adma.201600889.

[18] S. McVitie, D. McGrouther, S. McFadzean, D.A. MacLaren, K.J. O'Shea, M.J. Benitez, Aberration corrected Lorentz scanning transmission electron microscopy, Ultramicroscopy 152 (2015) 57–62, doi:10.1016/j.ultramic.2015.01.003.

[19] C. Phatak, A.K. Petford-Long, M. De Graef, Three-dimensional study of the vector potential of magnetic structures, Phys. Rev. Lett. 104 (25) (2010) 253901, doi:10.1103/PhysRevLett.104.253901.

[20] M.R. Teague, Deterministic phase retrieval: a Greens function solution, J. Opt. Soc. Am. 73 (11) (1983) 1434, doi:10.1364/JOSA.73.001434.

[21] Marc De Graef, Yimei Zhu, Magnetic Imaging and Its Applications to Materials, vol. 36, first ed., San Diego: Academic Press, San Diego, 2001.

[22] C.T. Koch, Towards full-resolution inline electron holography, Micron 63 (2014) 69–75, doi:10.1016/j.micron.2013.10.009.

[23] A. Budruk, C. Phatak, A.K. Petford-Long, M. De Graef, In situ Lorentz TEM magnetization studies on a Fe–Pd–Co martensitic alloy, Acta. Mater. 59 (17) (2011) 6646–6657, doi:10.1016/j.actamat.2011.07.020.

[24] A. Budruk, C. Phatak, A.K. Petford-Long, M. De Graef, In situ lorentz TEM magnetization study of a Ni–Mn–Ga ferromagnetic shape memory alloy, Acta. Mater. 59 (12) (2011) 4895–4906, doi:10.1016/j.actamat.2011.04.031.

[25] S.D. Pollard, V. Volkov, Y. Zhu, Propagation of magnetic charge monopoles and Dirac flux strings in an artificial spin-ice lattice, Phys. Rev. B 85 (18) (2012) 180402, doi:10.1103/PhysRevB.85.180402.

[26] S.D. Pollard, L. Huang, K.S. Buchanan, D.A. Arena, Y. Zhu, Direct dynamic imaging of non-adiabatic spin torque effects, Nat. Commun. 3 (2012) 1028, doi:10.1038/ncomms2025.

[27] H.S. Park, J.S. Baskin, A.H. Zewail, 4d Lorentz electron microscopy imaging: magnetic domain wall nucleation, reversal, and wave velocity, Nano Lett. 10 (9) (2010) 3796–3803, doi:10.1021/nl102861e.

[28] X.Z. Yu, N. Kanazawa, W.Z. Zhang, T. Nagai, T. Hara, K. Kimoto, Y. Matsui, Y. Onose, Y. Tokura, Skyrmion flow near room temperature in an ultralow current density, Nat. Commun. 3 (2012) 988, doi:10.1038/ncomms1990.

[29] C. Phatak, M. Tanase, A.K. Petford-Long, M. De Graef, Determination of magnetic vortex polarity from a single Lorentz Fresnel image, Ultramicroscopy 109 (3) (2009) 264–267, doi:10.1016/j.ultramic.2008.11.003.

[30] S.A. Eastwood, D.M. Paganin, A.C.Y. Liu, Automated phase retrieval of a single-material object using a single out-of-focus image, Opt. Lett. 36 (10) (2011) 1878, doi:10.1364/OL.36.001878.

[31] C.T. Koch, A flux-preserving non-linear inline holography reconstruction algorithm for partially coherent electrons, Ultramicroscopy 108 (2) (2008) 141–150, doi:10.1016/j.ultramic.2007.03.007.

[32] S.A. Montoya, S. Couture, J.J. Chess, J.C.T. Lee, N. Kent, D. Henze, S.K. Sinha, M.-Y. Im, S.D. Kevan, P. Fischer, B.J. McMorran, V. Lomakin, S. Roy, E.E. Fullerton, Tailoring magnetic energies to form dipole skyrmions and skyrmion lattices, Phys. Rev. B 95 (2017) 024415.

[33] A.J. Morgan, A.V. Martin, A.J. D'Alfonso, C.T. Putkunz, L.J. Allen, Direct exit-wave reconstruction from a single defocused image, Ultramicroscopy 111 (9–10) (2011) 1455–1460, doi:10.1016/j.ultramic.2011.07.005.

[34] A.V. Martin, L.J. Allen, Direct retrieval of a complex wave from its diffraction pattern, Opt. Commun. 281 (20) (2008) 5114–5121, doi:10.1016/j.optcom.2008.07.006.

[35] A.V. Martin, A.I. Bishop, D.M. Paganin, L.J. Allen, Practical implementation of a direct method for coherent diffractive imaging, Ultramicroscopy 111 (7) (2011) 777–781, doi:10.1016/j.ultramic.2010.10.003.

[36] Y. Aharonov, D. Bohm, Significance of electromagnetic potentials in the quantum theory, Phys. Rev. 115 (3) (1959) 485.

[37] S. Walton, K. Zeissler, W. Branford, S. Felton, MALTS: a tool to simulate Lorentz transmission electron microscopy from micromagnetic simulations, IEEE Trans. Magn. 49 (8) (2013) 4795–4800, doi:10.1109/TMAG.2013.2247410.

[38] M.D. Graef, Y. Zhu, Quantitative noninterferometric Lorentz microscopy, J. Appl. Phys. 89 (11) (2001) 7177–7179, doi:10.1063/1.1355337.

[39] D. Paganin, K.A. Nugent, Noninterferometric phase imaging with partially coherent light, Phys. Rev. Lett. 80 (12) (1998) 2586–2589, doi:10.1103/PhysRevLett.80.2586.

[40] T.E. Gureyev, C. Raven, A. Snigirev, I. Snigireva, S.W. Wilkins, Hard x-ray quantitative non-interferometric phase-contrast microscopy, J. Phys. D Appl. Phys. 32 (5) (1999) 563, doi:10.1088/0022-3727/32/5/010.

[41] V.V. Volkov, Y. Zhu, M. De Graef, A new symmetrized solution for phase retrieval using the transport of intensity equation, Micron 33 (5) (2002) 411–416, doi:10.1016/S0968-4328(02)00017-3.

[42] A. Lubk, G. Guzzinati, F. Borrnert, J. Verbeeck, Transport of intensity phase retrieval of arbitrary wave fields including vortices, Phys. Rev. Lett. 111 (17) (2013) 173902, doi:10.1103/PhysRevLett.111.173902.

[43] R. Chang, S. Li, M.V. Lubarda, B. Livshitz, V. Lomakin, FastMag: fast micromagnetic simulator for complex magnetic structures (invited), J. Appl. Phys. 109 (7) (2011) 07D358, doi:10.1063/1.3563081.

[44] M. Mansuripur, Computation of electron diffraction patterns in Lorentz electron microscopy of thin magnetic films, J. Appl. Phys. 69 (4) (1991) 2455–2464, doi:10.1063/1.348682.

[45] T.E. Gureyev, K.A. Nugent, Phase retrieval with the transport-of-intensity equation II Orthogonal series solution for nonuniform illumination, J. Opt. Soc. Am. A 13 (8) (1996) 1670, doi:10.1364/JOSAA.13.001670.

[46] T. Tasdizen, E. Jurrus, R.T. Whitaker, Non-uniform illumination correction in transmission electron microscopy, 2008, pp. 5–6.

[47] A. Feist, N. Bach, N.R. da Silva, T. Danz, M. Möer, K.E. Priebe, T. Domröse, J.G. Gatzmann, S. Rost, J. Schauss, S. Strauch, R. Bormann, M. Sivis, S. Schäfer, C. Ropers, Ultrafast transmission electron microscopy using a laser-driven field emitter: Femtosecond resolution with a high coherence electron beam, Ultramicroscopy, Available online 11 December 2016, ISSN 0304-3991, http://dx.doi.org/10.1016/j.ultramic.2016.12.005. (http://www.sciencedirect.com/science/article/pii/S0304399116303709).